

plication attacks, external hardware or removable media (such as USB drives), and compromised remote access are among the prevalent threats. Data theft and ransom-based attacks continue to plague organisations. Moreover, the proliferation of APIs (application programming interfaces) has opened new avenues for cybercriminals to exploit. In the realm of Industrial IoT, threat actors come in various forms. Negligent insiders, whether intentionally or unintentionally, remain a significant concern. Phishing attacks targeting employees, including top executives, are rampant. Cybercriminals driven by profit motives, rival companies seeking an edge, and malicious insiders coerced into compromising an organisation's security all play a role in the evolving threat landscape. Additionally, hacktivists, motivated by specific agendas, can wreak havoc, and nation-state actors pose a grave danger, especially in the context of geopolitical conflicts.

The industrial control system Kill Chain offers insight into the stages of a cyberattack in this domain. It starts with reconnaissance, where hackers gather information. Next comes weaponisation, where malware is crafted. The attackers then target specific individuals, often through social engineering tactics. Delivery and exploitation follow, with malware being introduced and exploited. Installation of malicious software within the OT network occurs before the attackers execute the attack, exfiltrate data, and compromise the control system.

Overcoming security challenges

While recognising the importance of security in Industrial IoT is a crucial first step, implementing effective measures can be challenging. Several factors contribute to the complexity of securing these environments:

Legacy systems. Many OT systems are built on legacy infrastructure, making it challenging to integrate modern security solutions seamlessly.

Lack of visibility. Managing and monitoring the vast number of assets in Industrial IoT environments can be overwhelming, leading to blind spots in security.

Compatibility issues. Existing security tools may not be compatible with OT systems, necessitating tailored solutions.

Security strategy. A well-defined security strategy, starting from the top down, is often lacking, resulting in inadequate protection.

Governance and monitoring. Consistent security governance and continuous monitoring are essential for safeguarding Industrial IoT environments, but these are often overlooked.

Lack of security as an end goal. Security should not be an afterthought but an integral part of the entire process. Neglecting security for the sake of convenience



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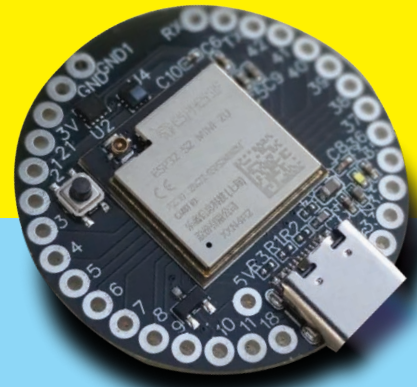
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